

PAPER_243 — NGC 3603 Full MUGE: Time-Varying Mass $M(t)$ and Additive Cavity Pressure $P(t)/\rho$

NGC 3603 Extreme Young Star Cluster — Complete 10-Term Master Universal Gravity Equation

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Framework: Unified Quantum Field Framework (UQFF) v4.10

Calculator Class: NGC3603FullMUGECavityPressureCalculator (CP3, Session 60)

Encoded By: Grok (xAI), October 2025 (C++ source); Python CP3 integration March 2026

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Abstract

This paper presents the complete 10-term Master Universal Gravity Equation (MUGE) for the NGC 3603 extreme young star cluster, incorporating two novel mathematical elements not previously captured in CP3:

1. **Time-varying mass** $M(t) = M_0(1 + \dot{M}_{\text{factor}} e^{-t/\tau_{SF}})$ — exponential star-formation inflow driving cluster mass growth over the first few Myr.
2. **Additive cavity pressure acceleration** $P(t)/\rho_{\text{fluid}}$ where $P(t) = P_0 e^{-t/\tau_{\text{exp}}}$ — the ionized cavity pressure from O/B-star UV/wind feedback expressed as an acceleration term alongside (not as a modifier of) the gravitational field.

These are **expressly distinct** from CP3 class 88 (NGC3603StellarPressureModulationCalculator, Session 55), which uses pressure as a multiplicative suppressor $g \propto (1 - P(t))$. The present paper treats pressure as an independent additive acceleration channel.

2. System Parameters

Parameter	Symbol	Default Value	Units
Initial mass	M_0	$4 \times 10^5 M_{\odot}$	kg
Cluster radius	r	9.5 ly	m
Star-formation timescale	τ_{SF}	1 Myr	s
Initial cavity pressure	P_0	4×10^{-8}	Pa
Pressure expansion timescale	τ_{exp}	1 Myr	s
Wind density	ρ_{wind}	10^{-20}	kg/m ³
Wind velocity	v_{wind}	2×10^6	m/s
Fluid density	ρ_{fluid}	10^{-20}	kg/m ³
Magnetic field	B	10^{-5}	T

3. Novel Mathematical Elements

3.1 Time-Varying Cluster Mass

$$M(t) = M_0(1 + \dot{M}_{\text{factor}} e^{-t/\tau_{SF}})$$

where $\dot{M}_{\text{factor}}$ is the dimensionless peak accretion enhancement. At $t = 0$, $M(0) = M_0(1 + \dot{M}_{\text{factor}})$ (maximum infall); at $t \gg \tau_{SF}$, $M \rightarrow M_0$ (steady state). The **star-formation rate** driving this is:

$$\frac{dM}{dt} = -\frac{M_0 \dot{M}_{\text{factor}}}{\tau_{SF}} e^{-t/\tau_{SF}}$$

The **star-formation efficiency** at time t is:

$$\varepsilon_{SF}(t) = \frac{M(t) - M_0}{M_0} = \dot{M}_{\text{factor}} e^{-t/\tau_{SF}}$$

3.2 Additive Cavity Pressure Acceleration

The O/B-star population inflates an ionized HII cavity. The cavity pressure decays exponentially as the molecular cloud is dispersed:

$$P(t) = P_0 e^{-t/\tau_{\text{exp}}}$$

This converts to a specific acceleration (per unit mass of ambient gas):

$$T_{\text{pressure}} = \frac{P(t)}{\rho_{\text{fluid}}} = \frac{P_0 e^{-t/\tau_{\text{exp}}}}{\rho_{\text{fluid}}}$$

Dispersal condition: The cluster disperses its natal cloud when the cavity-pressure acceleration exceeds local gravity:

$$\frac{P(t_{\text{disp}})}{\rho_{\text{fluid}}} = T_1 \Rightarrow t_{\text{disp}} = \tau_{\text{exp}} \ln\left(\frac{P_0}{\rho_{\text{fluid}} \cdot T_1}\right)$$

4. Full 10-Term MUGE

$$g_{\text{NGC3603}}(r, t) = T_1 + T_2 + T_3 + T_4 + T_q + T_{\text{fl}} + T_{\text{osc}} + T_{\text{DM}} + T_{\text{wind}} + T_{\text{pressure}}$$

Term 1 — Base with $M(t)$, Hubble, magnetic:

$$T_1 = \frac{GM(t)}{r^2}(1 + H_0 t)(1 - B/B_{\text{crit}})$$

Note: Unlike class 88 which applies $(1 - P)$ here, the full MUGE leaves T_1 unmodified and adds P explicitly as T_{10} .

Term 2 — UQFF unified field:

$$T_2 = (U_{g1}(t) + U_{g4}(t))(1 + f_{TRZ}), \quad U_{gk}(t) \propto M(t)/r^2$$

Term 3 — Cosmological constant:

$$T_3 = \frac{\Lambda c^2}{3}$$

Term 4 — EM with vacuum ratio:

$$T_4 = \frac{qv_{\text{gas}}B}{m_p} \left(1 + \frac{\rho_{\text{UA}}}{\rho_{\text{SCm}}} \right) s_{\text{EM}}$$

Term 5 — Quantum uncertainty:

$$T_q = \frac{\hbar}{\sqrt{\Delta x} \Delta p} \psi \frac{2\pi}{t_H}$$

Term 6 — Fluid acceleration:

$$T_{\text{fl}} = \frac{\rho_{\text{fluid}} V g_{\text{base}}(t)}{M(t)}$$

Term 7 — Two-mode oscillation:

$$T_{\text{osc}} = 2A \cos(kx) \cos(\omega t) + \frac{2\pi}{t_{\text{Gyr}}} A \cos(kx - \omega t)$$

Term 8 — Dark matter with $3GM(t)/r^3$ tidal perturbation:

$$T_{\text{DM}} = \frac{(M(t) + M_{\text{DM}})(\delta\rho/\rho + 3GM(t)/r^3)}{M(t)}$$

Term 9 — Stellar wind ram pressure:

$$T_{\text{wind}} = \frac{\rho_{\text{wind}} v_{\text{wind}}^2}{\rho_{\text{fluid}}}$$

Term 10 — Cavity pressure (additive, novel):

$$T_{\text{pressure}} = \frac{P_0 e^{-t/\tau_{\text{exp}}}}{\rho_{\text{fluid}}}$$

5. Distinction from Prior Art

Feature	Class 88 (Session 55)	This class (Session 60)
Mass	Fixed M	$M(t) = M_0(1 + \dot{M}e^{-t/\tau})$
Pressure form	Multiplicative $(1 - P(t))$ modifier	Additive $P(t)/\rho_{\text{fluid}}$ term
Physics	UV/wind pressure suppresses gravity	Cavity pressure drives independent acc.
Terms	4 (base, Ug, Λ , wind)	10 (all MUGE terms)
Quantum term	Not present	T_q present
Fluid term	Not present	T_{fl} present
2-mode oscillation	Not present	Standing + traveling wave

Feature	Class 88 (Session 55)	This class (Session 60)
DM pert2	Not present	$3GM(t)/r^3$

6. Cross-Validation: Class 88 vs Class 111 Regime

For small P_0 and short $t \ll \tau_{\text{exp}}$:

$$P(t) \approx P_0 \Rightarrow T_{\text{pressure}} \approx P_0/\rho_{\text{fl}}$$

For Class 88, the $(1 - P)$ effect on T_1 :

$$\Delta T_1 = -T_1^{\text{Class88}} \cdot P \approx -\frac{GM}{r^2}P$$

The two are **equivalent only in the limit** where $P_0/\rho_{\text{fl}} = (GM/r^2)P$, i.e., when $P_0 = GM\rho_{\text{fl}}/r^2$. At default values of NGC 3603, this equality does not hold — the additive and multiplicative forms predict **different total** g and different dispersal timescales.

7. Numerical Example

At $t = 0.5$ Myr, default parameters:

$$M_{\text{dot}}(t) = 1.0 \times e^{-0.5} \approx 0.607 \Rightarrow M(t) \approx 1.607 M_0$$

$$P(t) = 4 \times 10^{-8} \times e^{-0.5} \approx 2.43 \times 10^{-8} \text{ Pa}$$

$$T_{\text{pressure}} = \frac{2.43 \times 10^{-8}}{10^{-20}} = 2.43 \times 10^{12} \text{ m/s}^2$$

The cavity pressure acceleration dominates all other terms at early times — consistent with the observational picture of NGC 3603's OB-star cluster rapidly dispersing its natal cloud within ~ 3 Myr (Harayama et al. 2008; Pang et al. 2013).

8. Simulation Recommendations

1. τ_{SF} **sweep** (0.5–5 Myr): Track $M(t)$ peak mass vs. cluster binding energy
 2. P_0 **sweep** (10^{-9} – 10^{-6} Pa): Determine cavity pressure dispersal threshold
 3. **Term10 vs Term9 cross-over**: Find t^* where $T_{\text{pressure}}(t^*) = T_{\text{wind}}$
 4. M_{dot} **sweep** (0.1–2.0): Map star-formation efficiency vs. gravitational retention
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9. Integration Into UQFF Pipeline

- **CP3 class:** NGC3603FullMUGECavityPressureCalculator (113th AST node, 112th calculator, Session 60)
 - **Aggregator:** v2.4.0 — registered in CP3_CALCULATORS
 - **Source:** Doc 11 C++ class NGC3603 (Grok/xAI, October 2025)
 - **Complementary class:**
 - Class 88 NGC3603StellarPressureModulationCalculator — multiplicative pressure form (Session 55)
 - Class 110 RingsOfRelativityEinsteinLensingMUGECalculator — companion new class (PAPER_242)
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10. References

- Harayama Y. et al. (2008): ApJ 675, 1319 — NGC 3603 stellar mass function
 - Pang X. et al. (2013): ApJ 764, 73 — star formation history
 - Portegies Zwart S. et al. (2010): Nature 464, 203 — young massive clusters
 - Murphy, D.T. (2025): UQFF Manuscript — Doc 11 (NGC 3603 MUGE, October 2025)
 - Grok (xAI): C++ class NGC3603, encoded October 2025
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*PAPER_243 | Session 60 | CP3 class 112 (NGC3603FullMUGECavityPressureCalculator)
| UQFF v4.10*